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A record of the Lebanese Lizard (*Phoenicolacerta laevis*) in northern Türkiye: Evidence of possible human-mediated dispersal

Mehmet Kürşat Şahin^a, Ortaç Çetintaş^b, Alexey Yanchukov^b, Muammer Kurnaz^{c,*}

^aDepartment of Biology, Faculty of Science, Hacettepe University, Ankara, Türkiye;

^bDepartment of Biology, Faculty of Science, Zonguldak Bülent Ecevit University, Zonguldak, Türkiye; ^cDepartment of Medical Services and Techniques, Kelkit Sema Doğan Vocational

School of Health Services, Gümüşhane University, Kelkit, Türkiye

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The Lebanese Lizard (*Phoenicolacerta laevis*), primarily native to the Levant region, has recently been discovered in isolated populations well beyond its known distribution. This study presents the first confirmed record of *P. laevis* along the Black Sea coast of northern Türkiye, specifically in the city of Karadeniz Ereğli, Zonguldak Province. Mitochondrial cytochrome *b* (cyt *b*) sequence analyses revealed a high degree of genetic similarity (>99%) between the Karadeniz Ereğli specimens and those from both Andırın, Kahramanmaraş (within the species' native range in southern Türkiye) and Anaklia on Georgia's Black Sea coast, suggesting a shared origin. We propose that the presence of *P. laevis* in Karadeniz Ereğli is likely the result of human-mediated introduction, possibly via maritime trade, as the lizards were found restricted to the walls of a single building near a major port. A comparable introduction scenario has previously been suggested for the Anaklia population. The highly localized occurrence and absence of the species in neighboring areas indicate a recent introduction; however, the factors inhibiting further spread within the urban landscape remain unclear. This finding contributes to the broader understanding of species distribution dynamics and underscores the significant role of anthropogenic activity in facilitating the range expansion of non-native reptiles.

Keywords: Distribution; introduced species; Black Sea; cytochrome *b*; Zonguldak

Introduction

The expansion of terrestrial reptile species and their invasions present a complex interplay of ecological dynamics, anthropogenic influences, and biogeographical patterns. Terrestrial reptiles, which encompass a diverse array of species, are increasingly facing threats from habitat destruction, climate change, and invasive species, which can significantly alter their distribution and community structures (Cox et al., 2022). One of the primary drivers of reptile range expansion and introductions is habitat alteration due to human activities. Agriculture, urbanization, and hunting are pervasive threats that have been documented to impact reptile populations globally (Farooq et al., 2024). For instance, in regions where agricultural practices dominate, the native reptile assemblages often experience declines, while invasive species may thrive in modified environments (Hawllitschek et al., 2011). This phenomenon is for example evident in the Comoro Islands, where habitat degradation has favoured invasive species, leading to competition and predation pressures on endemic reptiles (Hawllitschek et al., 2011). The ongoing trend of increasing reptile invasions is particularly pronounced in islands and

*Corresponding author. Email: muammerkurnazz@gmail.com

Table 1. *Phoenicolacerta* and *Podarcis* specimens used in phylogenetic tree with their GenBank accession numbers and locations.

Species	Locality	Acc. Number	References
<i>P. laevis</i>	Karadeniz Ereğli/Türkiye	PV101162	This study
<i>P. laevis</i>	Çamlıyayla/Mersin/Türkiye	JN673207	Pavlicev et al. (unpubl.)
<i>P. laevis</i>	Anaklia/Georgia	MF438266	Tarkhnishvili et al., 2017
<i>P. laevis</i>	Damascus/Syria	JN673208	Pavlicev et al. (unpubl.)
<i>P. laevis</i>	Jerusalem/Israel	JN673209	Pavlicev et al. (unpubl.)
<i>P. laevis</i>	Bludan/Syria	JN673210	Pavlicev et al. (unpubl.)
<i>P. laevis</i>	Andırın/K. Maraş/Türkiye	JN673211	Pavlicev et al. (unpubl.)
<i>P. laevis</i>	Harbiye/Hatay/Türkiye	JN673212	Pavlicev et al. (unpubl.)
<i>P. laevis</i>	Samandağ/Hatay/Türkiye	JN673213	Pavlicev et al. (unpubl.)
<i>P. laevis</i>	Andırın/K. Maraş/Türkiye	DQ461745	Pavlicev & Mayer (2006)
<i>P. laevis</i>	Yarpuz/Adana/Türkiye	DQ461744	Pavlicev & Mayer (2006)
<i>P. laevis</i>	Harbiye/Hatay/Türkiye	DQ461747	Pavlicev & Mayer (2006)
<i>P. laevis</i>	Titus Tunnel/Hatay/Türkiye	MN015197	Garcia-Porta et al. (2019)
<i>P. laevis</i>	Juniyah/Lebanon	DQ461756	Pavlicev & Mayer (2006)
<i>P. laevis</i>	Byblos/Lebanon	DQ461761	Pavlicev & Mayer (2006)
<i>P. laevis</i>	Craq Chevalier/Syria	DQ461752	Pavlicev & Mayer (2006)
<i>P. cyanisparsa</i>	Al Barah/Syria	JN673216	Pavlicev et al. (unpubl.)
<i>P. cyanisparsa</i>	Al Barah/Syria	DQ461750	Pavlicev & Mayer (2006)
<i>P. cyanisparsa</i>	Gaziantep/Türkiye	DQ461748	Pavlicev & Mayer (2006)
<i>P. cyanisparsa</i>	Gaziantep/Türkiye	DQ461749	Pavlicev & Mayer (2006)
<i>P. troodica</i>	Cyprus	MN015173	Garcia-Porta et al. (2019)
<i>P. kulzeri</i>	Maalula/Syria	DQ461763	Pavlicev & Mayer (2006)
<i>P. kulzeri</i>	Maalula/Syria	DQ461764	Pavlicev & Mayer (2006)
<i>Podarcis muralis</i>	Dinslaken/Germany	HQ652881	Schulte et al. (2008)

economically developed regions, where the ecological impacts of these invaders can be profound (Capinha et al., 2017).

The Levantine rock lizards of the genus *Phoenicolacerta* Arnold, Arribas & Carranza, 2007 include four species distributed across the Levant and Mediterranean regions: , *P. kulzeri*, *P. cyanisparsa*, and *P. troodica* (Arnold et al., 2007; Sindaco & Jeremčenko, 2008; Tamar et al., 2015). Although no representatives of the genus were previously known to occur outside the Levant and Mediterranean, Tarkhnishvili et al. (2017) reported the presence of the Lebanon Lizard (), the most widespread species of the genus and native to southern Türkiye, Lebanon, northern Israel, Jordan, and western Syria, in Anaklia, Georgia.

In the present study, we report the occurrence of *P. laevis* along the Black Sea coast of northern Türkiye, confirmed through both visual identification and molecular DNA analysis. We also assess whether this represents an autochthonous population or a recent introduction.

Material and Methods

The animals were caught by hand. The clipped tips of tails obtained were kept in 96% ethanol at -20°C. After tissue samples were taken, the lizards were released back into their natural habitat. No voucher samples were taken, the tissues were stored in Hacettepe University Herpetology Laboratory (HU-HERP-T007-009).



Figure 1. Urban habitat of *Phoenicolacerta laevis* locality in Karadeniz Ereğli.

Total genomic DNA was extracted from the tail tissue sample using a DNAeasy extraction kit following the manufacturer's protocol (AMBRD Laboratories, Istanbul, Türkiye). The partial *cyt b* gene (526 bp) was amplified and sequenced using the primers L15369 (Forward) and H15915 (Reverse) (Gabelaia et al., 2015). The PCR protocol for *cyt b* included: 3 min initial denaturation of 93°C, 60-sec denaturation of 93°C, 53-sec annealing of 60°C, and 120-sec extension of 69°C, for 30 cycles, and a final 10 min extension of 70°C. After the PCR step, we purified the PCR products with an appropriate kit following the manufacturer's protocol (AMBRD Laboratories, Istanbul, Türkiye). The purified PCR products were outsourced for sequencing on a Sanger platform to Macrogen-Europe Inc.

Phylogenetic analyses were based on *cyt b* gene sequences obtained from our material, and additional 526 base pair sequences of *Phoenicolacerta* species from their native range and from Georgia retrieved from GenBank. The corresponding single haplotype sequence (acc. no. PV101162) was used in the phylogenetic tree and uploaded to GenBank. All *cyt b* sequences used in the molecular analysis were aligned by using ClustalW method in the software Geneious Prime 2019. The best-fit substitution model with the lowest AIC (Akaike's Information Criterion) value (Akaike, 1974) was determined in JModelTest v.2.1.8 (Darriba et al. 2012). The GTR+I was chosen as the best-fit base substitution model. To reconstruct the Species ID Tree, we carried out Maximum Likelihood (ML) analysis in MEGA 12 (Kumar et al., 2024), using a heuristic search method (10,000 random addition replicates tree-bisection-reconnection, TBR, branch swapping), and bootstrap analyses with 1000 replications (Felsenstein, 1985). HQ652881 *Podarcis muralis* (Schulte et al., 2007) sequence for *cyt b* gene was obtained from Gene Bank and used as an out-group in the analyses. Uncorrected pairwise (P distance) sequence divergences for the *cyt b* gene were calculated in MEGA 12 (Kumar et al., 2024).

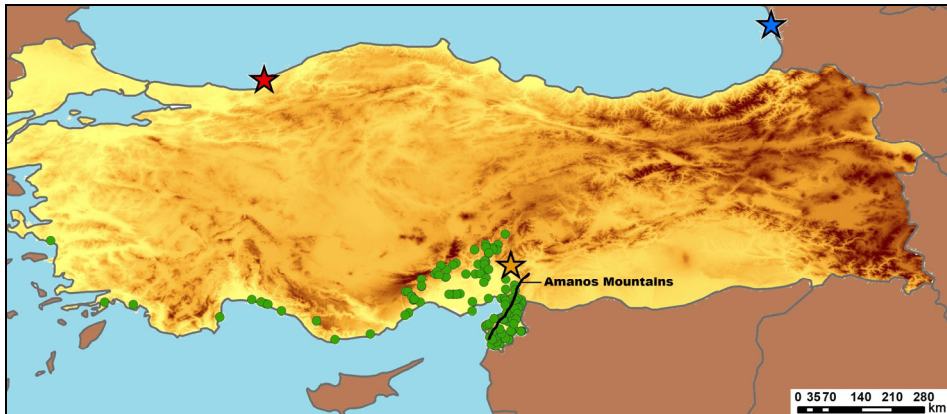


Figure 2. Distribution of *Phoenicolacerta laevis* in Türkiye. Green circles: literature records; red star: new localities from northern Türkiye; blue star: previous record from Georgia. The yellow star shows Andırın-Kahramanmaraş. The black line shows Amanos Mountains.

Results and Discussion

Three individuals of *Phoenicolacerta laevis* (1♂, 1♀, 1 subadult ♀) were collected within the town of Karadeniz Ereğli, Zonguldak Province on the Eastern Black Sea (41.295°N 31.414°E). They were found on the garden walls of a single house located approximately 1000 m from the present-day port of Karadeniz Ereğli. The habitat is approximately 100 m a.s.l. and is confined to the side walls and garden walls of single house (Figure 1). The species lives in sympatry with other lizard species in the area: *Podarcis siculus* and *Lacerta viridis*. The occurrence of both sexes was confirmed by body colouration and additionally the presence of hemipenis in the male. Males possess brighter colours (orange, yellow, blue) than females.

In the genetic analysis, no variable position was determined in the partial base sequences of the 526-base *cyt b* gene belonging to three different *P. laevis* samples. A total of 146 variable nucleotide positions were identified among 23 partial *cyt b* gene sequences (each approximately 526 bp in length) retrieved from GenBank for phylogenetic comparison (Table 1). While most clades received high bootstrap support (>90), a few internal nodes showed relatively lower support, indicating well-supported relationships among the analyzed *Phoenicolacerta* populations. The resulting species ID phylogenetic tree showed that *P. laevis* can be divided into two different lineages (Figure 3). One branch is represented by *P. laevis* individuals from south of Amanos Mountain mentioned in Tamar et al. (2015), while the other branch including *P. laevis* from Türkiye as well as *P. troodica*, and *P. cyanisparsa* species. The samples from Karadeniz Ereğli cluster with those from Andırın, Kahramanmaraş (Southern Türkiye), and the Anaklia (Black Sea coast, Georgia). The latter population is thought to have arrived there from the Levantine-Mediterranean region via human-mediated dispersal (Tarkhnishvili et al., 2017). The genetic distance between Karadeniz Ereğli and Anaklia was 0.2% (Table 2).

While the Lebanese lizard is distributed in the Levantine, the Mediterranean and the southern part of the Aegean region (Karış & Göçmen, 2014; Ilgaz et al., 2016). Our record is the first from Turkish Black Sea coast and complements a record in Georgia

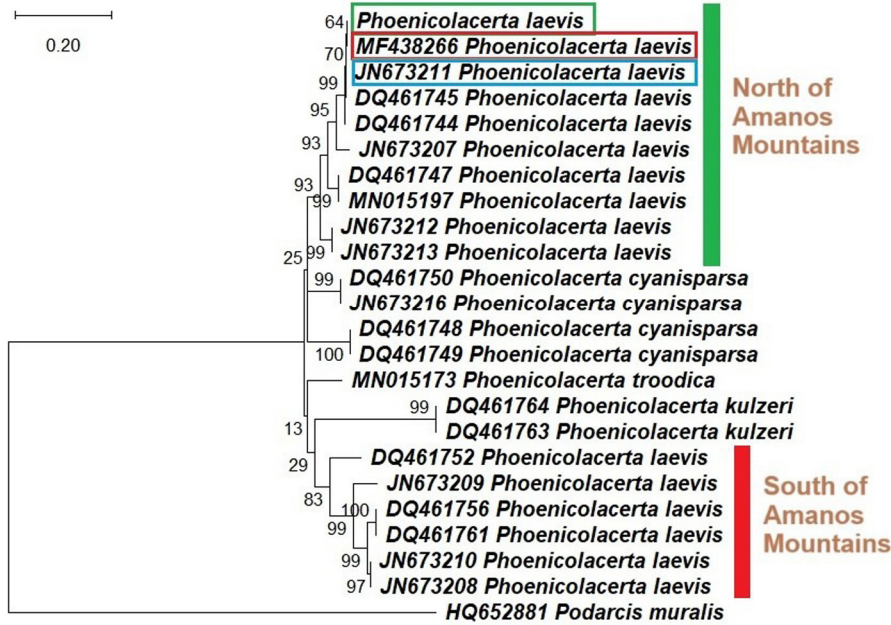


Figure 3. Species Identification Tree based on 526 bp *cyt b* gene sequence of the mitochondrial DNA. Bootstrap values given by ML. The green rectangle shows the Karadeniz Ereğli specimen, the red rectangle the Anaklia (Georgia) specimen and the blue rectangle shows the Andırın specimen from southern Turkey.

Table 2. Uncorrected genetic distances for partial sequences of *cyt b* gene in *Phoenicolacerta laevis*. The accession number of the specimen from the Turkish Black Sea coast is shown in **bold**.

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	PV101162																
2	JN673211	0.006															
3	MF438266	0.002	0.002														
4	DQ461745	0.006	0.000	0.002													
5	DQ461744	0.008	0.006	0.006	0.006												
6	JN673207	0.050	0.048	0.048	0.048	0.046											
7	DQ461747	0.058	0.056	0.056	0.056	0.054	0.064										
8	MN015197	0.058	0.056	0.056	0.056	0.054	0.064	0.004									
9	JN673212	0.071	0.065	0.067	0.065	0.069	0.077	0.060	0.060								
10	JN673213	0.071	0.065	0.067	0.065	0.069	0.077	0.060	0.060	0.000							
11	DQ461756	0.141	0.144	0.144	0.144	0.139	0.130	0.133	0.128	0.133	0.133						
12	DQ461761	0.146	0.149	0.149	0.149	0.144	0.135	0.138	0.133	0.138	0.138	0.004					
13	JN673210	0.149	0.152	0.152	0.152	0.147	0.142	0.136	0.131	0.145	0.145	0.025	0.025				
14	JN673208	0.152	0.154	0.155	0.154	0.150	0.145	0.139	0.134	0.148	0.148	0.027	0.027	0.002			
15	DQ461752	0.150	0.148	0.151	0.148	0.148	0.156	0.144	0.144	0.135	0.135	0.130	0.130	0.113	0.113		
16	JN673209	0.169	0.161	0.162	0.161	0.162	0.180	0.165	0.165	0.155	0.155	0.092	0.092	0.079	0.082	0.130	

on the eastern Black Sea coast (Tarkhnishvili et al., 2017). Karadeniz Ereğli is a large industrial and port city and it has been an important maritime trade region since antiquity. It is and has historically been well-connected by marine traffic to the south of Türkiye, especially with the large commercial ports such as Mersin and Iskenderun. The fact that the population in Karadeniz Ereğli is genetically very close to Andırın in southern Turkey may suggest its origin from the native range of the Lebanese lizard, perhaps

accidentally brought by humans via shipping of goods. The same scenario has been proposed for the origin of the Anaklia population in Georgia (Tarkhnishvili et al. 2017), which is genetically almost identical to Karadeniz Ereğli.

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Disclosure Statement

No potential conflict of interest was reported by the authors.

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